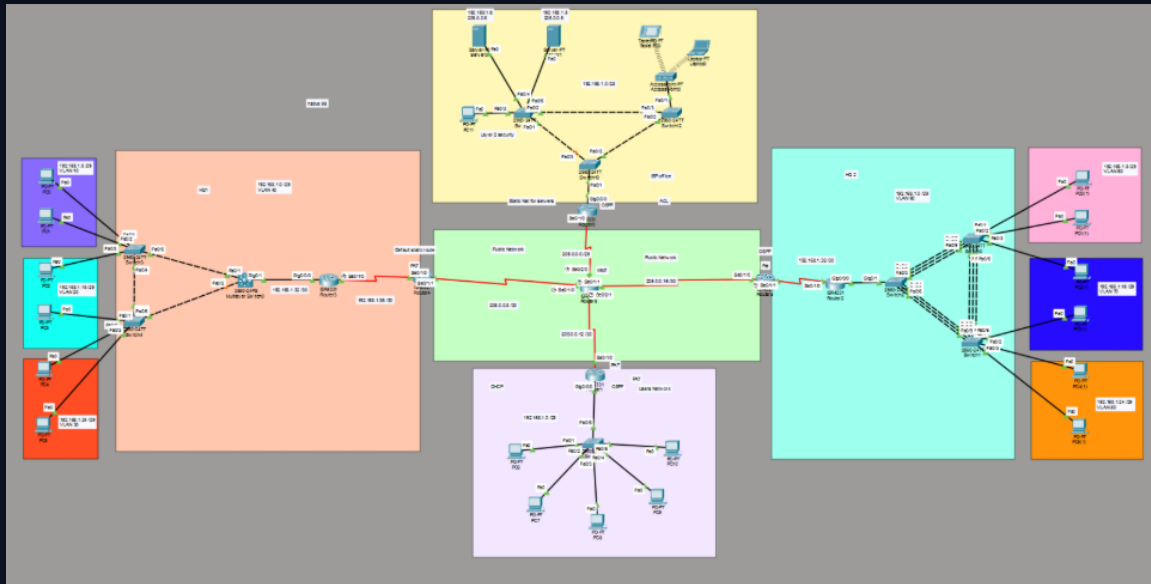


ENTERPRISE NETWORK ARCHITECTURE SPECIFICATION

Supply Chain Network

7 Routers • 1 Multi-Layer Switch (MLS) • 9 Switches • 22 Endpoints

OSPF Routing, VLANs, LACP EtherChannel, DHCP Pools, Dynamic PAT, Static NAT & Remote Control



STUDENT NAME

Moaaz Mosaad Mosaad

Network Design & Implementation

PROJECT SUPERVISOR

Eng. Saher Walid

Supervisor / Instructor

Project Executive Summary & Architecture Roadmap

This document outlines the complete technical specifications, configuration baselines, and verification procedures for the multi-branch enterprise supply chain network infrastructure. Below is the master architecture roadmap detailing all 10 implementation parts.

Part	Section / Module Title	Key Technologies	Primary Scope / Objective
Part 1	Project Goals	Topology Design, Security Scope	Secure end-to-end communication to public servers without leaking private IPv4 addresses
Part 2	Subnetting & Device Tables	VLSM IPv4 (/26, /27, /28, /30)	Comprehensive IP addressing plan across HQ1, HQ2, Users, ISP Office, and Public Servers
Part 3	End-Device Addressing	Static IP, DHCP Pools, Exclusions	Deterministic server/gateway addressing and dynamic lease allocation for user endpoints
Part 4	VLANs & EtherChannel (LACP)	802.1Q Trunks, Channel-Groups	Layer 2 broadcast domain isolation and switch-to-switch redundant trunk link bundling
Part 5	MLS Routing & Gateways	Inter-VLAN SVI, Router-on-a-Stick	Core collapsed backbone routing, default gateway redundancy, and routed uplinks
Part 6	OSPF Routing & Default Route	OSPFv2 (Area 0), Static Route	Interior dynamic routing mesh convergence and quad-zero (0.0.0.0/0) public ISP route propagation
Part 7	PAT (NAT Overload)	NAT Overload, Standard ACLs	Internal private host internet access (HQ1, HQ2, Users) via single outside routable public IP
Part 8	Static NAT for Servers	One-to-One Static NAT	Bi-directional public exposure for internal servers without exposing private server addresses
Part 9	Security & Remote Control	SSH, Telnet	Secure terminal management across all switches and routers using local authentication
Part 10	Testing & Verification Matrix	ICMPv4	End-to-end multi-protocol test cases, diagnostic show commands, and issue resolution logs

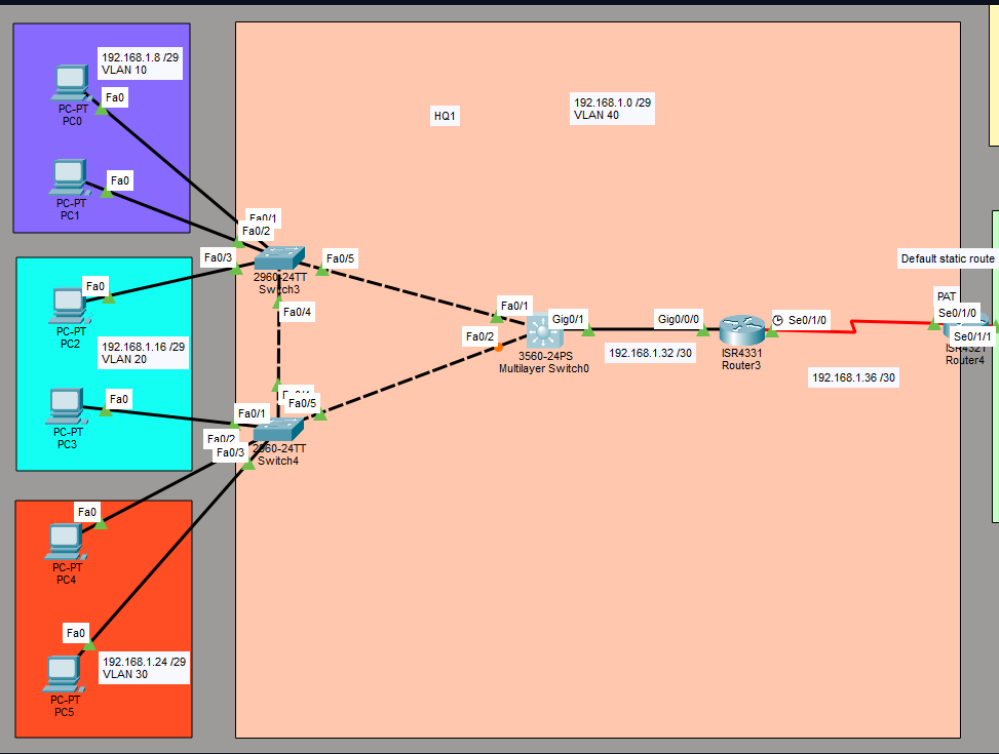
Part 1: Project Goals

This project aims to enable End-devices to connect to servers on a public network without leaking any private IP address outside of private networks.

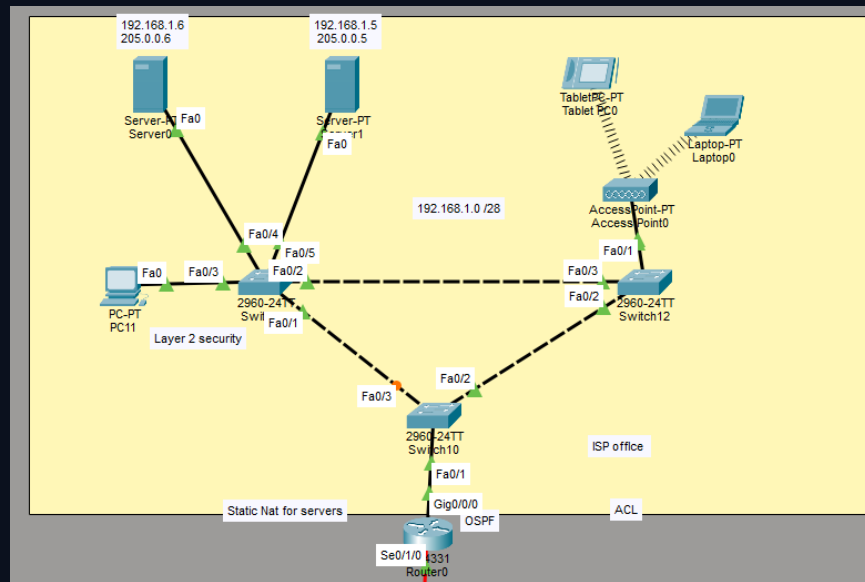
Part 2: VLSM Subnetting Architecture & Device Addressing Tables

Here is the subnetting table for each network in the project:

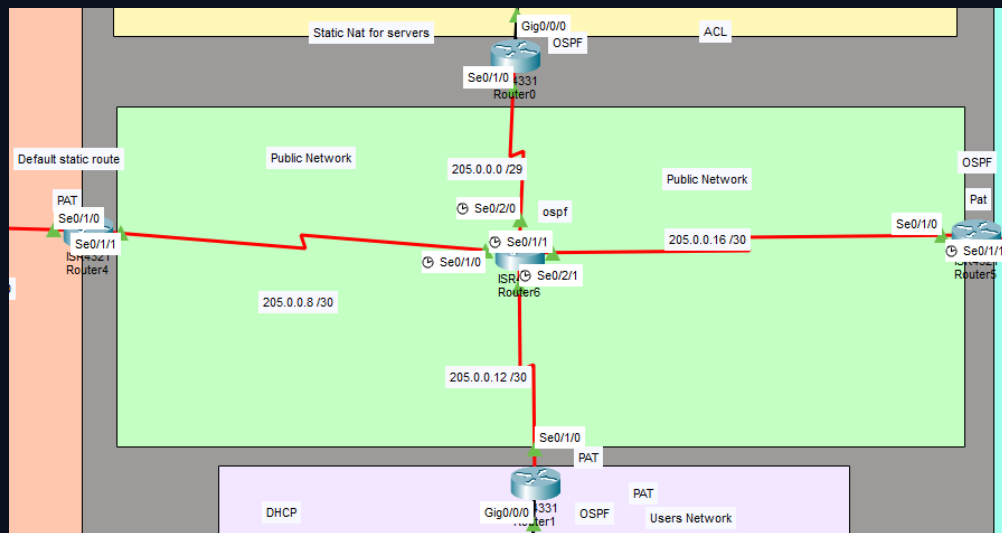
Network / Segment	Subnet Address	Public NAT / Gateway IP	Usable Host Range
HQ1 Department	192.168.1.0	205.0.0.10	.1 to .38
HQ2 Department	192.168.1.0	205.0.0.18	.1 to .34
Users Network	192.168.1.0/28	205.0.0.14	.1 to .14
ISP office	192.168.1.0/28	205.0.0.5, 205.0.0.6	.1 to .14
Public Network	205.0.0.0	all	.1 to .18



HQ1 Topology Segment (MLS, Switches, VLANs)



ISP Office Infrastructure



Public Network Core Router & WAN Mesh

Part 3: End-Device Addressing: Static IP & Dynamic DHCP Pool Configuration

Assigning the static IP address for the end device by going to Desktop -> IP configuration:

The screenshot shows the configuration window for PC0. The 'Desktop' tab is selected. The 'IP Configuration' section is expanded, showing the 'FastEthernet0' interface. Under 'IP Configuration', the 'Static' radio button is selected. The fields are filled with: IPv4 Address: 192.168.1.10, Subnet Mask: 255.255.255.248, Default Gateway: 192.168.1.9, and DNS Server: 0.0.0.0. The 'IPv6 Configuration' section is also expanded, showing the 'Static' radio button selected. The fields are: IPv6 Address: (empty), Link Local Address: FE80::290:21FF:FE84:162C, Default Gateway: (empty), and DNS Server: (empty). The '802.1X' section is expanded, showing 'Use 802.1X Security' checked, Authentication: MD5, Username: (empty), and Password: (empty). A 'Top' button is at the bottom left.

Static IP Configuration on PC0

DHCP Configuration

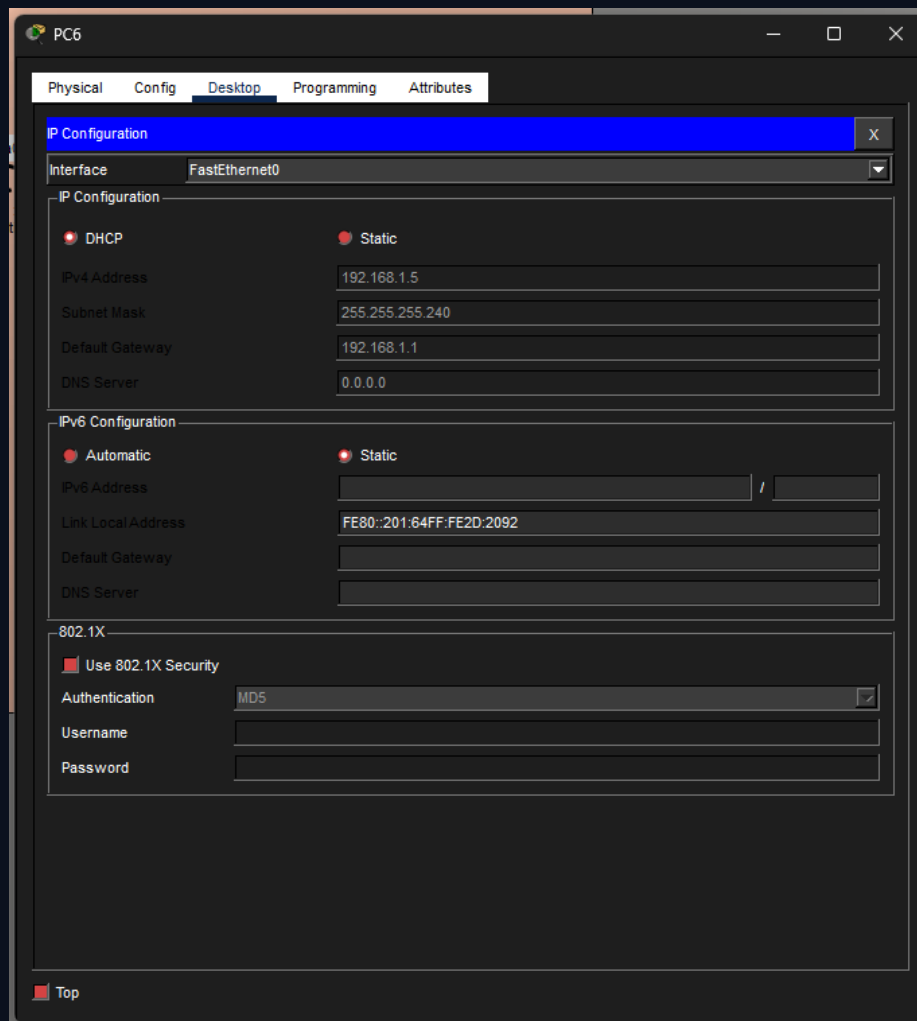
Then Making the DHCP Pool after excluding the router and switch's IPs from the network:

```
ip dhcp excluded-address 192.168.1.1 192.168.1.2
!
ip dhcp pool 1
 network 192.168.1.0 255.255.255.240
 default-router 192.168.1.1
```

DHCP Pool Commands

DHCP Lease Verification

And then Assigning the DHCP IPs for the Users network end devices:



DHCP Verification on PC6

Part 4: Layer 2 Segmentation: VLANs, 802.1Q Trunks & EtherChannel (LACP/PAgP)

Configure the VLANs on the HQ2 switches:

```
S(config)#Vlan 60
S(config-vlan)#Vlan 70
S(config-vlan)#Vlan 80
S(config-vlan)#Vlan 90
S(config-vlan)#Vlan 99
```

VLAN Creation Commands

Access Ports

Then configure the links with the end devices as access with the correct VLAN:

```
S(config-vlan)#exit
S(config)#int range f0/1-2
S(config-if-range)#sw
S(config-if-range)#switchport m
S(config-if-range)#switchport mode ac
S(config-if-range)#switchport mode access
S(config-if-range)#sw
S(config-if-range)#switchport ac
S(config-if-range)#switchport access v
S(config-if-range)#switchport access vlan 60
S(config-if-range)#int f0/3
S(config-if)#sw
S(config-if)#switchport m
S(config-if)#switchport mode ac
S(config-if)#switchport mode access
S(config-if)#sw
S(config-if)#switchport ac
S(config-if)#switchport access v
S(config-if)#switchport access vlan 70
```

Access Ports Configuration

EtherChannel Bundling

Then configure the ether channels between the switches:

```

S(config-if)#exit
S(config)#int range f0/4-6
S(config-if-range)#channel-gr
S(config-if-range)#channel-group 1 mode ac
S(config-if-range)#channel-group 1 mode active
S(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-3-UPDOWN: Interface Port-channel1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface Port-channel1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up
S(config-if-range)#port-channel 1
^
% Invalid input detected at '^' marker.

S(config-if-range)#int port-channel 1
S(config-if)#sw
S(config-if)#switchport m
S(config-if)#switchport mode tr
S(config-if)#switchport mode trunk
S(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-3-UPDOWN: Interface Port-channel1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface Port-channel1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

S(config-if)#sw
S(config-if)#switchport tr
S(config-if)#switchport trunk al
S(config-if)#switchport trunk allowed v
S(config-if)#switchport trunk allowed vlan 60,70,80,90
S(config-if)#sw
S(config-if)#switchport t
S(config-if)#switchport trunk n
S(config-if)#switchport trunk native v
S(config-if)#switchport trunk native vlan 99
S(config-if)#

```

EtherChannel Configuration

Part 5: Multi-Layer Switch (MLS) Routing & Default Gateway Configuration

Configure the VLANs on the HQ1 MLS:

```

S(config)#Vlan 10
S(config-vlan)#Vlan 20
S(config-vlan)#Vlan 30
S(config-vlan)#Vlan 40
S(config-vlan)#Vlan 99
S(config-vlan)#exit

```

MLS VLANs

SVI Gateways

Then assigning the Gate-way IP for each VLAN:

```
S(config)#int vlan 10
S(config-if)#ip add 192.168.1.9 255.255.255.248
S(config-if)#no shut
```

SVI Gateways

HQ2 Sub-interfaces

Also making sub interfaces on the HQ2 gateway to assign the Gate-way IP for each VLAN:

```
R1(config)#int g0/0/0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int g0/0/0.60
R1(config-subif)#enc
R1(config-subif)#encapsulation dot1Q 60
R1(config-subif)#ip add 192.168.1.9 255.255.255.248
R1(config-subif)#exit
R1(config)#
```

Router-on-a-Stick Sub-Interfaces

Part 6: Enterprise Dynamic Routing (OSPF) & Public Default Static Routing

Configuring Dynamic routing (OSPF) on the private routers in HQ 1, 2 and the MLS in HQ1:

```
S(config)#ip routing
S(config)#router ospf 1
S(config-router)#netwo
S(config-router)#network 192.168.1.8 0.0.0.7 area 0
S(config-router)#network 192.168.1.16 0.0.0.7 area 0
S(config-router)#network 192.168.1.24 0.0.0.7 area 0
S(config-router)#network 192.168.1.0 0.0.0.7 area 0
S(config-router)#network 192.168.1.32 0.0.0.3 area 0
S(config-router)#exit
S(config)#int g0/1
S(config-if)#sw
S(config-if)#no switchport
S(config-if)#end
```

MLS OSPF Configuration

```
R(config)#router ospf 1
R(config-router)#net
R(config-router)#network 192.168.1.32 0.0.0.3 area 0
R(config-router)#network 192.168.1.36 0.0.0.3 area 0
R(config-router)#end
```

Router OSPF Configuration

Default Static Routing

Configure the default static routing on the edge routers to the public networks:

```

R(config)#ip route 0.0.0.0 0.0.0.0 s0/1/1
R(config)#router ospf 1
R(config-router)#net
R(config-router)#network 0.0.0.0 255.255.255.255 area 0
R(config-router)#def
R(config-router)#default-information
R(config-router)#default-information originate
R(config-router)#end

```

Edge Default Route Injection

Core Public Router OSPF

Dynamic routing for the core-public router too:

```

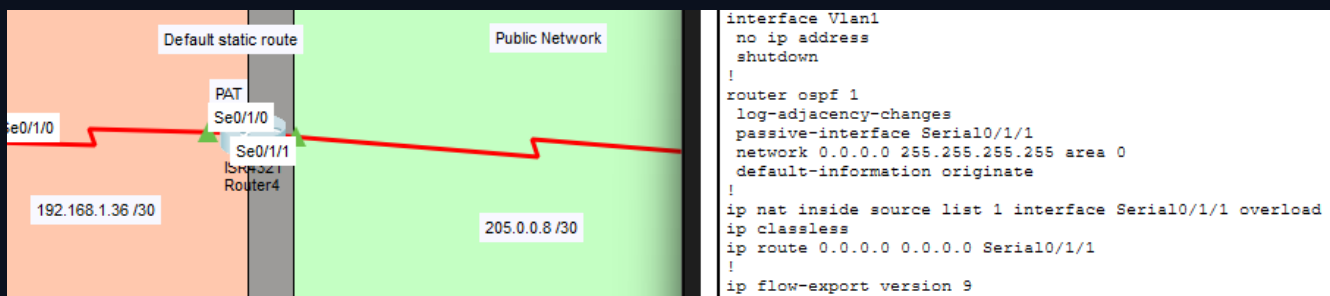
R_Root(config)#router ospf 1
R_Root(config-router)#network 205.0.0.0 0.0.0.7 area 0
R_Root(config-router)#network 205.0.0.8 0.0.0.3 area 0
R_Root(config-router)#network 205.0.0.12 0.0.0.3 area 0
R_Root(config-router)#network 205.0.0.16 0.0.0.3 area 0
R_Root(config-router)#end

```

Core Public Router OSPF

Passive-Interface

Making the public interfaces of the edge routers passive:



Edge Router Passive-interface

Part 7: Port Address Translation (PAT / NAT Overload)

Configure the PAT for the HQ 1, 2 and Users network's edge router:

```

R(config)#acc
R(config)#access-list 1 per
R(config)#access-list 1 permit 192.168.1.0 0.0.0.255
R(config)#ip na
R(config)#ip nat in
R(config)#ip nat inside s
R(config)#ip nat inside source 1
R(config)#ip nat inside source list 1 in
R(config)#ip nat inside source list 1 interface s0/1/1 ov
R(config)#ip nat inside source list 1 interface s0/1/1 overload
R(config)#end

```

PAT Configuration

Inside & Outside Interfaces

Then configure the inside and outside interfaces:

```

R(config)#int s0/1/0
R(config-if)#ip nat ins
R(config-if)#ip nat inside
R(config-if)#int s0/1/1
R(config-if)#ip nat outs
R(config-if)#ip nat outside
R(config-if)#end

```

NAT Inside/Outside

Part 8: Static NAT for Servers

Configure a static NAT for the servers so that each private IP has exactly one public IP:

```

R(config)#ip nat ins
R(config)#ip nat inside
R(config)#ip nat inside source st
R(config)#ip nat inside source static 192.168.1.5 205.0.0.5
R(config)#ip nat inside source static 192.168.1.6 205.0.0.6
R(config)#int g0/0/0
R(config-if)#ip nat inside
R(config-if)#int s0/1/0
R(config-if)#ip nat out
R(config-if)#ip nat outside
R(config-if)#end

```

Static NAT Mapping

Core Router Routing to Servers

Now let's make a static routing on the core-public router so that it knows where the servers are located:

```

R_Root(config)#ip route 205.0.0.5 255.255.255.255 s0/2/0
R_Root(config)#ip route 205.0.0.6 255.255.255.255 s0/2/0
R_Root(config)#end

```

Static Route to Servers

Part 9: Security & Remote Control

Configure SSH and Telnet with crypto key on switches and routers:

```

Switch#
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#ena
Switch(config)#enable sec
Switch(config)#enable secret cisco
Switch(config)#line con 0
Switch(config-line)#pass
Switch(config-line)#password cisco
Switch(config-line)#login
Switch(config-line)#exi
Switch(config)#ser
Switch(config)#service p
Switch(config)#service password-encryption
Switch(config)#ho S
S(config)#ip domain an
S(config)#ip domain name cisco.com
S(config)#cr k g r g m 1024
The name for the keys will be: S.cisco.com

% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 1:42:21.406: %SSH-5-ENABLED: SSH 1.99 has been enabled
S(config)#username cisco secret cisco
S(config)#line vty 0 3
S(config-line)#login local
S(config-line)#trans input all
S(config-line)#end
S#
%SYS-5-CONFIG_I: Configured from console by console
cop
S#copy r
S#copy running-config st
S#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
S#ex

```

SSH & Telnet Hardening

Part 10: End-to-End Verification, Connectivity Testing & Troubleshooting Matrix

Testing the VLANs routing and then the basic dynamic routing up to the server's public IP:

```
PC0
Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.26

Pinging 192.168.1.26 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.26: bytes=32 time<1ms TTL=127
Reply from 192.168.1.26: bytes=32 time<1ms TTL=127
Reply from 192.168.1.26: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.26:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.33

Pinging 192.168.1.33 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.33: bytes=32 time<1ms TTL=254
Reply from 192.168.1.33: bytes=32 time=1ms TTL=254
Reply from 192.168.1.33: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.1.33:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>192.168.1.37
Invalid Command.

C:\>ping 192.168.1.37

Pinging 192.168.1.37 with 32 bytes of data:

Reply from 192.168.1.37: bytes=32 time=1ms TTL=253
Reply from 192.168.1.37: bytes=32 time=3ms TTL=253
Reply from 192.168.1.37: bytes=32 time=1ms TTL=253
Reply from 192.168.1.37: bytes=32 time=2ms TTL=253

Ping statistics for 192.168.1.37:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\>ping 205.0.0.9

Pinging 205.0.0.9 with 32 bytes of data:

Reply from 205.0.0.9: bytes=32 time=23ms TTL=252
Reply from 205.0.0.9: bytes=32 time=4ms TTL=252
Reply from 205.0.0.9: bytes=32 time=4ms TTL=252
Reply from 205.0.0.9: bytes=32 time=4ms TTL=252

Ping statistics for 205.0.0.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 23ms, Average = 8ms

C:\>ping 205.0.0.5

Pinging 205.0.0.5 with 32 bytes of data:

Request timed out.
Reply from 205.0.0.5: bytes=32 time=7ms TTL=123
Reply from 205.0.0.5: bytes=32 time=6ms TTL=123
Reply from 205.0.0.5: bytes=32 time=6ms TTL=123

Ping statistics for 205.0.0.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 7ms, Average = 6ms

C:\>|
```

Ping Verification Tests

PAT Verification

And made sure that the PC is getting out to the public network with a public IP which is the edge router's public IP:

```
R#show ip nat tr
Pro Inside global      Inside local      Outside local     Outside global
icmp 205.0.0.5:21       192.168.1.5:21   205.0.0.10:21    205.0.0.10:21
icmp 205.0.0.5:22       192.168.1.5:22   205.0.0.10:22    205.0.0.10:22
icmp 205.0.0.5:23       192.168.1.5:23   205.0.0.10:23    205.0.0.10:23
--- 205.0.0.5          192.168.1.5      ---              ---
--- 205.0.0.6          192.168.1.6      ---              ---
```

Show IP NAT Translations Output

SSH Verification

Testing the SSH and Telnet remote access:

```
S#ssh -l cisco 192.168.1.33

Password:

R>exit

[Connection to 192.168.1.33 closed by foreign host]
```

Remote SSH Verification